Origin–destination matrix recovery from partially observed equilibrium link flows

Восстановление матрицы корреспонденций по частично наблюдаемым равновесным потокам

Overall assessment

The manuscript addresses an important inverse problem in transportation modelling and, in my view, contains a potentially publishable contribution. The combination of Beckmann equilibrium assignment, equilibrium sensitivity, entropy regularization, and IPF enforcement of known marginals is coherent, and I especially appreciate that the authors evaluate not only the residual on observed links but also the full-network flow error and performance after a local topology change. My comments below are mainly intended to clarify what the method adds beyond existing work and to make the numerical evidence easier to interpret.

Main comments and questions

1. **Novelty and comparison with existing methods.** The literature review is relevant and includes classical OD estimation, bilevel equilibrium formulations, equilibrium sensitivity, modern differentiation approaches, and sensor-location studies. At the same time, the central sensitivity idea is clearly related to earlier work, especially Josefsson–Patriksson (2007), while Patwary et al. (2023), Liu et al. (2023), and Guarda et al. (2024) provide alternative ways to obtain gradients through or around the equilibrium problem. Could the authors state more explicitly what they regard as the main methodological novelty here: the sparse shortest-path-subgraph realization, the reuse of one factorization for many OD perturbations, the integration with exact marginals through IPF, or the complete recovery framework? A short comparison of these aspects would help the reader judge the contribution more precisely.

   I also wonder whether at least one numerical baseline could be added. Sioux Falls is a standard benchmark for OD estimation, yet the manuscript compares sensor masks only within the proposed method. However, a baseline run under exactly the same protocol would be very informative. Could the authors provide a modest numerical validation or comparison of the proposed Jacobian calculation against finite differences for several feasible OD perturbations? If computationally reasonable, comparison with another equilibrium-aware gradient approach under the same experimental setting would also help quantify the practical benefit of the proposed sensitivity construction.

2. **Initialization and sensitivity to the prior OD matrix.**

   If I understand Eq. (5.1) correctly, the initial estimate is generated as

   $$D_0 = \mathrm{IPF}_{l,w}\left(P_{\mathrm{off}}\left[\max(D_{\mathrm{true}} + \Xi, \varepsilon)\right]\right), \qquad \Xi_{ij} \sim \mathcal{N}(0, \sigma_0^2), \qquad \sigma_0 = 1.6 D_+.$$

   The reported relative L_1 error of D_0 is 0.663357, so this is clearly not a trivially accurate initial estimate. I also think that starting from an available prior OD matrix while imposing known production and attraction totals is quite reasonable from a transportation-planning perspective. Nevertheless, because the particular D_0 used here is constructed directly by perturbing D_{true}, I wonder how dependent the Adam–IPF procedure is on the structure of this starting point.

   Would the authors consider one additional run from a structurally different but feasible initialization, for example a maximum-entropy or gravity-type matrix satisfying the same marginals? I would view such a test not as a requirement to start from an unrealistic uninformative prior, but rather as a simple indication of whether the quality of the returned flow-consistent solution is sensitive to the structure of the initial OD estimate. This seems particularly interesting because a relatively large initial OD error already corresponds to a much smaller full-network flow error of approximately 0.114, which itself illustrates the weak identifiability discussed in the paper.

3. **Interpretation of "OD-matrix recovery".**

   The manuscript explicitly states in Section 3.3 that the objective does not contain information about D_{true} and is not intended to minimize $\|D_{\mathrm{true}} - D^*\|$. I understand and appreciate this point: under partial observations the OD matrix is weakly identifiable, and reproduction of equilibrium network flows is therefore a more relevant objective than proximity to one particular synthetic ground-truth matrix.

   Russian is not my native language, so it is possible that I have missed some nuance in the terminology; nevertheless, I was slightly uncertain about the intended meaning of "recovered

OD matrix" in this setting. Is D^* best interpreted as one *flow-consistent OD estimate* among potentially many matrices capable of producing similar equilibrium flows? If so, perhaps this interpretation could be stated somewhat more directly near the formulation and in the conclusion.

Final assessment

Overall, I consider the manuscript publishable after moderate modifications. The core formulation appears sound, and the requested changes mainly concern clarification and modest numerical validation.